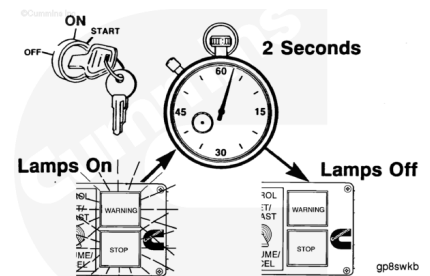


Trainings ▾

Initial Check

Turn the keyswitch to the ON position while monitoring the fault lamps. The fault lamps **must** illuminate for 2 to 3 seconds.

If lamps do **not** illuminate, check for burned-out bulbs.

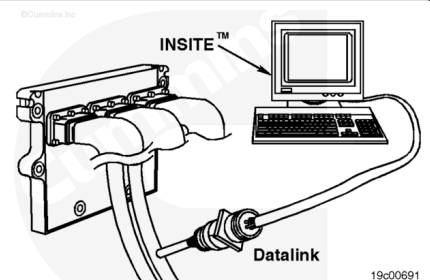


Turn the keyswitch to the OFF position.

Connect an electronic service tool to the vehicle datalink.

Turn the keyswitch to the ON position.

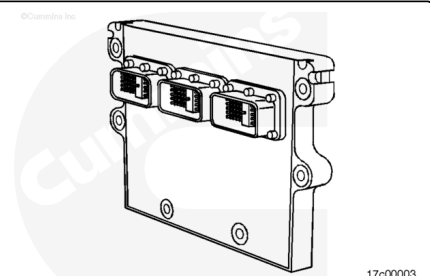
Select the monitor mode on the electronic service tool. The electronic service tool **must** be able to communicate with ECM. If the ECM will **not** communicate with the service tool, see the Communication Error - Electronic Service Tool or Control Device symptom tree.



Remove

⚠ CAUTION ⚠

Record all programmable parameters, features, and calibration information from the old ECM before disconnecting the harness connectors. This information will be needed to program the new ECM.

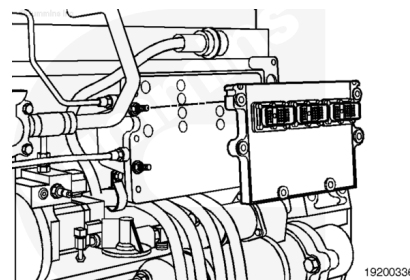


See the INSITE™ Manual under 'Save as a Template' for information on how to electronically save and restore ECM parameters.

Disconnect the actuator, sensor, and OEM harness connectors from the ECM, if they are **not** already removed.

Remove the metric capscrews that secure the ECM to the cooling plate on the engine block.

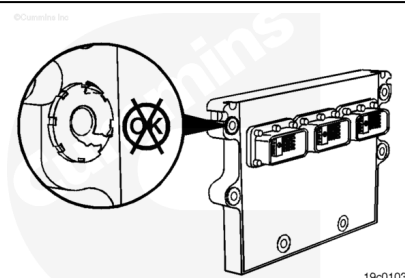
Remove the ECM from the cooling plate.



Install

Make certain that the star washer is in place on the ECM upper-left mounting hole.

The star washer **must** be free of cracks or damage that will prevent a proper ground connection to the engine block.



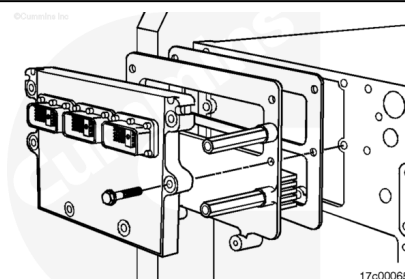
⚠ CAUTION ⚠

Do not paint the backside of the ECM. Make sure no grease or dirt is between the ECM and the engine block.

Install the new ECM to the cooling plate.

Tighten the capscrews.

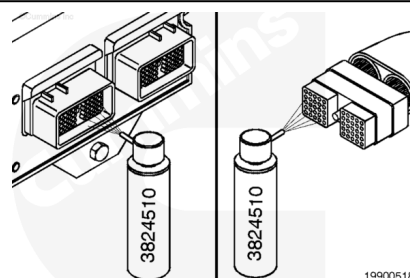
Torque Value: 20 n•m [177 in-lb]



⚠ CAUTION ⚠

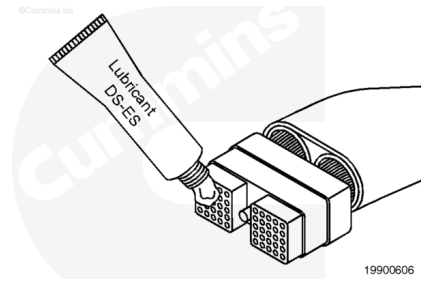
Do not blow compressed air into the ECM ports or connectors. Compressed air can contain moisture due to condensation.

Use quick-dry electrical contact cleaner, Part Number 3824510, to remove all dirt and moisture from the ECM connector ports and the harness connectors.



⚠ CAUTION ⚠

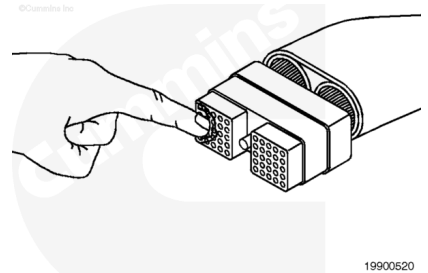
Use only Cummins-recommended lubricant DS-ES, Part Number 3822934. Other lubricants, such as lubricating oil or grease, in the connectors can cause ECM damage, poor engine performance, or premature connector wear.



Apply a thin coating of lubricant to the nosepiece of the connectors.

Spread the lubricant across the nosepiece so it enters every pin hole and lubricates the contacts.

Lubricant **must not** be visible on the surface of the nose piece.



Connect the actuator, sensor, and OEM harness connectors to the ECM ports.

Tighten the connector capscrews to the ECM.

Torque Value: 2.8 n·m [25 in-lb]

Note : When an ECM is replaced, the new ECM must be calibrated. Refer to Procedure [019-032 \(/qs3/pubsys2/xml/en/procedures/105/105-019-032.html\)](/qs3/pubsys2/xml/en/procedures/105/105-019-032.html).

